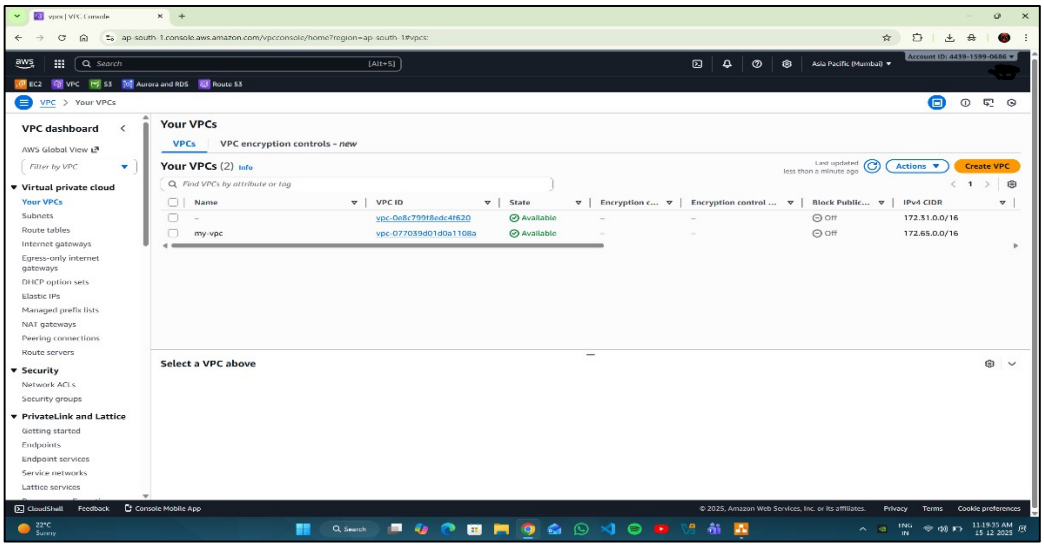
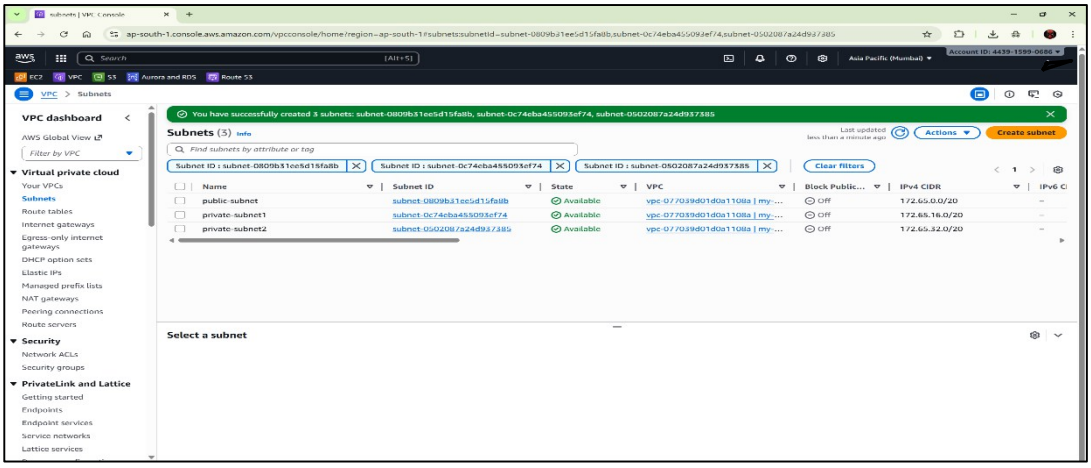
Title: 3-Tier project (Tomcat & MySQL)

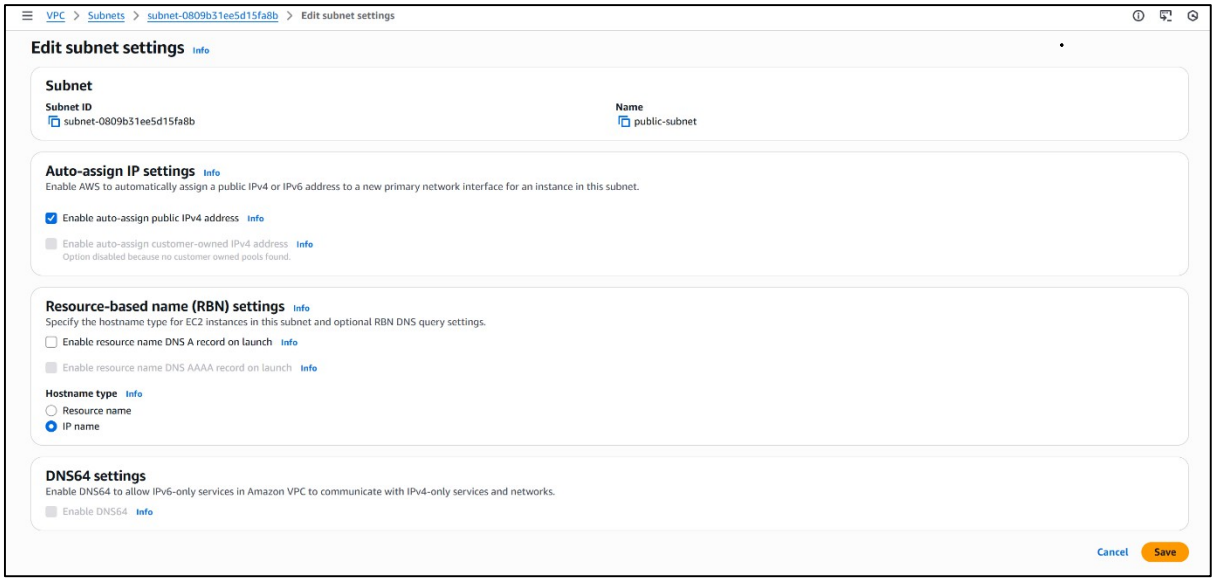
Step 1: Create VPC –



Step 2: Create 3 subnets of this VPC **(public, private, private-2)** –

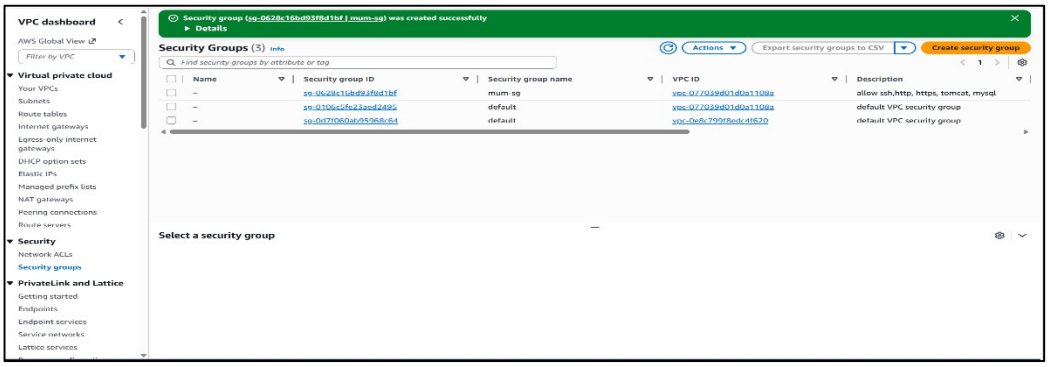


Step 3: Make a public subnet - In **Edit subnet setting** –

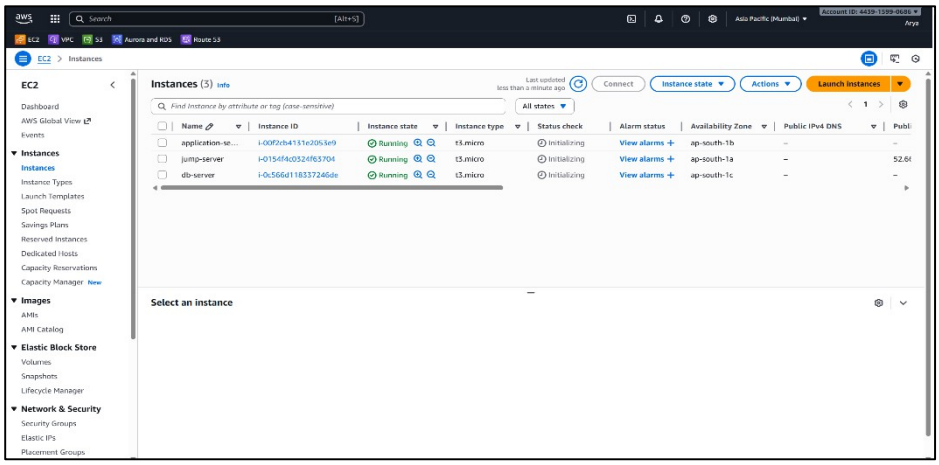


Title: 3-Tier project (Tomcat & MySQL)

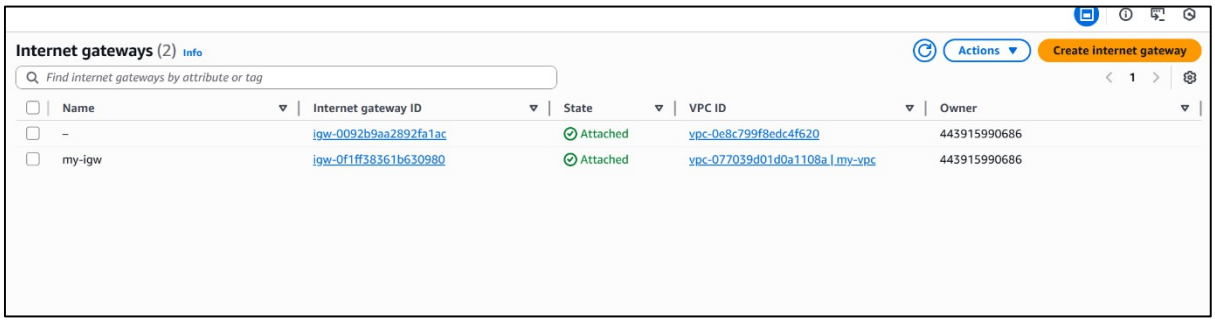
Step 4: Now create a Security Group – Allow SSH, Tomcat, MySQL, HTTP, HTTPS.



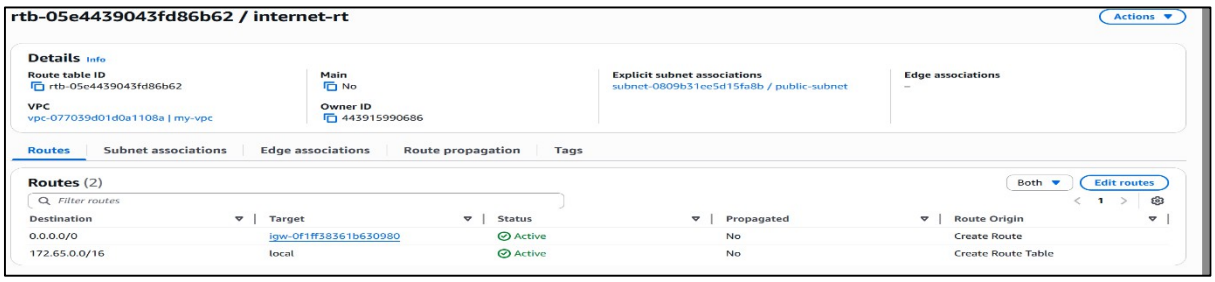
Step 5: Create instances – jump server in Public subnet, application server in private-subnet1 and Database server in private-subnet2. And we created a Security Group.



Step 6: Now create an Internet Gateway – It will give Internet to the VPC.



Step 7: Now create a Route table, and Route the Internet Gateway, and associate the Subnets with public only.

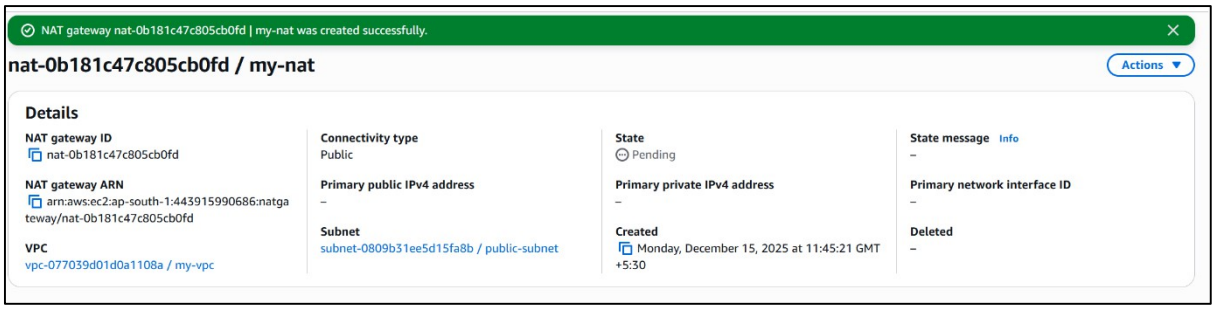


Title: 3-Tier project (Tomcat & MySQL)

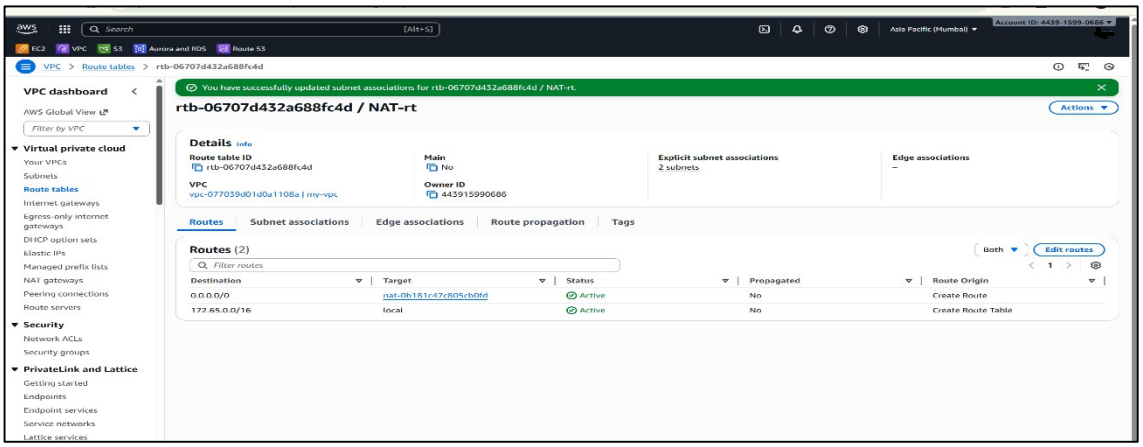
Step 8: Transfer the Key to the Jump server to get access to the application server

```
MINGW64:/c/Users/Lenovo/Downloads

Lenovo@DESKTOP-4ONKFG8 MINGW64 ~
$ cd Downloads/

Lenovo@DESKTOP-4ONKFG8 MINGW64 ~/Downloads
$ scp -i mumbai-key.pem mumbai-key.pem ec2-user@52.66.200.135:/home/ec2-user
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
mumbai-key.pem                                                    100% 1674    16.1KB/s   00:00

Lenovo@DESKTOP-4ONKFG8 MINGW64 ~/Downloads
$
```

Step 9: After accessing the Application server needs internet so create a NAT Gateway in the public subnet.



Step 10: After this, create a Route table for the NAT gateway associated with private subnets – Route NAT Gateway and Associate Subnets.



Step 11: Access the jump server and change the key permissions, and get access to the Application server and download Apache Tomcat in the application server.

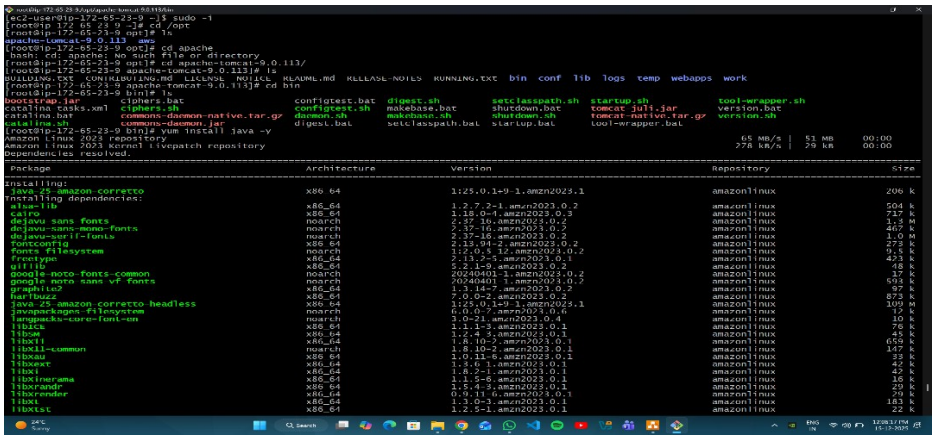
```
[ec2-user@ip-172-65-13-201 ~]$
[ec2-user@ip-172-65-13-201 ~]$ ls
mumbai-key.pem
[ec2-user@ip-172-65-13-201 ~]$ chmod 400 mumbai-key.pem
[ec2-user@ip-172-65-13-201 ~]$ ll
total 4
-r--------. 1 ec2-user ec2-user 1674 Dec 15 06:14 mumbai-key.pem
[ec2-user@ip-172-65-13-201 ~]$ ssh -i mumbai-key.pem ec2-user@172.65.23.9
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-65-23-9 ~]$ curl -O https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.113/bin/apache-tomcat-9.0.113.tar.gz
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 12.4M  100 12.4M    0     0   256M      0 --:--:-- --:--:-- --:--:--  259M
[ec2-user@ip-172-65-23-9 ~]$ ls
apache-tomcat-9.0.113.tar.gz
```

Title: 3-Tier project (Tomcat & MySQL)

Step 12: Extract this file to the/opt folder.

```
[ec2-user@ip-172-65-23-9 ~]$ sudo tar -xzvf apache-tomcat-9.0.113.tar.gz -C /opt
apache-tomcat-9.0.113/conf/
apache-tomcat-9.0.113/conf/catalina.policy
apache-tomcat-9.0.113/conf/catalina.properties
apache-tomcat-9.0.113/conf/context.xml
apache-tomcat-9.0.113/conf/jaspic-providers.xml
apache-tomcat-9.0.113/conf/jaspic-providers.xsd
apache-tomcat-9.0.113/conf/logging.properties
apache-tomcat-9.0.113/conf/server.xml
apache-tomcat-9.0.113/conf/tomcat-users.xml
apache-tomcat-9.0.113/conf/tomcat-users.xsd
apache-tomcat-9.0.113/conf/web.xml
apache-tomcat-9.0.113/bin/
apache-tomcat-9.0.113/lib/
apache-tomcat-9.0.113/logs/
apache-tomcat-9.0.113/temp/
apache-tomcat-9.0.113/webapps/
apache-tomcat-9.0.113/webapps/ROOT/
apache-tomcat-9.0.113/webapps/ROOT/WEB-INF/
apache-tomcat-9.0.113/webapps/docs/
apache-tomcat-9.0.113/webapps/docs/META-INF/
apache-tomcat-9.0.113/webapps/docs/WEB-INF/
apache-tomcat-9.0.113/webapps/docs/WEB-INF/jsp/
apache-tomcat-9.0.113/webapps/docs/annotationapi/
apache-tomcat-9.0.113/webapps/docs/api/
apache-tomcat-9.0.113/webapps/docs/appdev/
apache-tomcat-9.0.113/webapps/docs/appdev/sample/
apache-tomcat-9.0.113/webapps/docs/appdev/sample/docs/
apache-tomcat-9.0.113/webapps/docs/appdev/sample/src/
apache-tomcat-9.0.113/webapps/docs/appdev/sample/src/mypackage/
apache-tomcat-9.0.113/webapps/docs/appdev/sample/web/
apache-tomcat-9.0.113/webapps/docs/appdev/sample/web/WEB-INF/
apache-tomcat-9.0.113/webapps/docs/appdev/sample/web/images/
apache-tomcat-9.0.113/webapps/docs/architecture/
apache-tomcat-9.0.113/webapps/docs/architecture/requestProcess/
apache-tomcat-9.0.113/webapps/docs/architecture/startup/
apache-tomcat-9.0.113/webapps/docs/config/
apache-tomcat-9.0.113/webapps/docs/elapi/
apache-tomcat-9.0.113/webapps/docs/images/
apache-tomcat-9.0.113/webapps/docs/images/fonts/
apache-tomcat-9.0.113/webapps/docs/jaspicapi/
apache-tomcat-9.0.113/webapps/docs/jspapi/
```

Step 13: Go to **/opt/apache-tomcat-9.0.113/** and in that **bin/** here you see catalina.sh, so we have to start Tomcat with this file, **but Tomcat needs Java to start, so first install Java.**



Title: 3-Tier project (Tomcat & MySQL)

Step 14: Now start Tomcat - **./catalina.sh start**

```
root@ip-172-65-23-9:/opt/apache-tomcat-9.0.113/bin
[root@ip-172-65-23-9 bin]# ./catalina.sh start
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.113
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.113
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.113/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.113/bin/bootstrap.jar:/opt/apache-tomcat-9.0.113/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-65-23-9 bin]#
```

Step 15: Now we have to dump the website in the application server –

so go to - /opt/apache-tomcat-9.0.113/webapps/

```
[root@ip-172-65-23-9 apache-tomcat-9.0.113]# cd webapps/
[root@ip-172-65-23-9 webapps]# ls
ROOT  docs  examples  host-manager  manager
[root@ip-172-65-23-9 webapps]# curl -O https://s3-us-west-2.amazonaws.com/studentapi-cit/student.war
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 89423  100 89423    0     0  71525      0  0:00:01  0:00:01 --:--:-- 71538
[root@ip-172-65-23-9 webapps]# |
```

Step 16: Now everything is set, but we have to give the location to the jump server. We need nginx, so install nginx in the jump server.

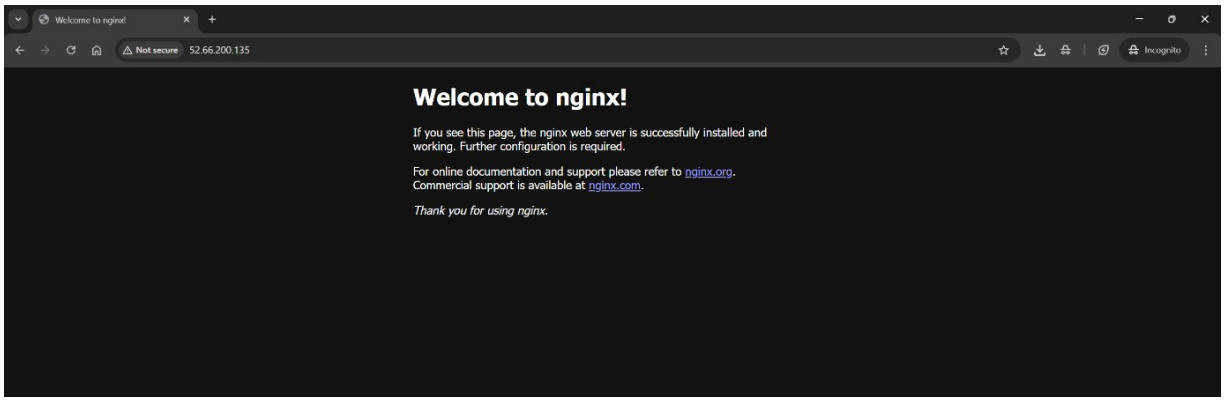
```
root@ip-172-65-13-201:~
[root@ip-172-65-13-201 ~]# yum install nginx -y
Last metadata expiration check: 0:00:18 ago on Mon Dec 15 06:22:12 2025.
Dependencies resolved.
================================================================================================================================
 Package                                  Architecture                Version                                  Repository                                  Size
================================================================================================================================
Installing:
 nginx                                    x86_64                      1:1.28.0-1.amzn2023.0.2                  amazonlinux                                 33 k
Installing dependencies:
 generic-logos-httpd                      noarch                      18.0.0-12.amzn2023.0.3                   amazonlinux                                 19 k
 gperftools-libs                          x86_64                      2.9.1-1.amzn2023.0.3                     amazonlinux                                108 k
 libunwind                                x86_64                      1.4.0-5.amzn2023.0.3                     amazonlinux                                 68 k
 nginx-core                               x86_64                      1:1.28.0-1.amzn2023.0.2                  amazonlinux                                686 k
 nginx-filesystem                         noarch                      1:1.28.0-1.amzn2023.0.2                  amazonlinux                                9.0 k
 nginx-mimetypes                          noarch                      2.1.49-3.amzn2023.0.3                    amazonlinux                                 21 k

Transaction Summary
================================================================================================================================
Install  7 Packages

Total download size: 1.1 M
Installed size: 3.7 M
Downloading Packages:
(1/7): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm                                                609 kB/s |  19 kB     00:00
(2/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm                                                            1.7 MB/s |  66 kB     00:00
(3/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm                                                      6.3 MB/s | 108 kB     00:00
(4/7): nginx-1.28.0-1.amzn2023.0.2.x86_64.rpm                                                               1.7 MB/s |  33 kB     00:00
(5/7): nginx-filesystem-1.28.0-1.amzn2023.0.2.noarch.rpm                                                    510 kB/s | 9.6 kB     00:00
(6/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm                                                     932 kB/s |  21 kB     00:00
(7/7): nginx-core-1.28.0-1.amzn2023.0.2.x86_64.rpm                                                           16 MB/s | 686 kB     00:00
--------------------------------------------------------------------------------------------------------------------------------
Total                                                                                                        9.7 MB/s | 1.1 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                                                                                                                    1/1
  Running scriptlet: nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch                                                           1/7
  Installing       : nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch                                                           1/7
  Installing       : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch                                                              2/7
  Installing       : libunwind-1.4.0-5.amzn2023.0.3.x86_64                                                                     3/7
  Installing       : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64                                                               4/7
  Installing       : nginx-core-1:1.28.0-1.amzn2023.0.2.x86_64                                                                 5/7
  Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                                                         6/7
  Installing       : nginx-1:1.28.0-1.amzn2023.0.2.x86_64                                                                      7/7
  Running scriptlet: nginx-1:1.28.0-1.amzn2023.0.2.x86_64                                                                      7/7
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                                                         1/7
  Verifying        : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64                                                               2/7
  Verifying        : libunwind-1.4.0-5.amzn2023.0.3.x86_64                                                                     3/7
  Verifying        : nginx-1:1.28.0-1.amzn2023.0.2.x86_64                                                                      4/7
  Verifying        : nginx-core-1:1.28.0-1.amzn2023.0.2.x86_64                                                                 5/7
  Verifying        : nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch                                                           6/7
  Verifying        : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch                                                              7/7

Installed:
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch    gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64    libunwind-1.4.0-5.amzn2023.0.3.x86_64    nginx-1:1.28.0-1.amzn2023.0.2.x86_64
  nginx-core-1:1.28.0-1.amzn2023.0.2.x86_64            nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch    nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
[root@ip-172-65-13-201 ~]# |
```

Step 17: Start nginx

```
[ec2-user@ip-172-65-13-201 ~]$ sudo -i
[root@ip-172-65-13-201 ~]# systemctl start nginx.service
[root@ip-172-65-13-201 ~]# systemctl enable nginx.service
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
[root@ip-172-65-13-201 ~]#
```

Step 18: After this, if you hit the public IP Address of the jump server, you see the nginx page.

Title: 3-Tier project (Tomcat & MySQL)



So after this, we have to add the location of the application server to see the website.

Step 19: First, log in root of the application server and go to - **/etc/nginx/.** Here, you see nginx.conf, so in this file we have to add the location.

```
[root@ip-172-65-13-201 ~]# cd /etc/nginx/
[root@ip-172-65-13-201 nginx]# ls
conf.d        fastcgi.conf.default    koi-utf     mime.types.default  scgi_params         uwsgi_params.default
default.d     fastcgi_params          koi-win     nginx.conf          scgi_params.default  win-utf
fastcgi.conf  fastcgi_params.default  mime.types  nginx.conf.default  uwsgi_params
[root@ip-172-65-13-201 nginx]# vim nginx.conf
[root@ip-172-65-13-201 nginx]# |
```

```
        root         /usr/share/nginx/html;

        # Load configuration files for the default server block.
        include /etc/nginx/default.d/*.conf;

        error_page 404 /404.html;
        location = /404.html {
        }

        location / {
        proxy_pass http://172.65.23.9:8080/student/;
        }

        error_page 500 502 503 504 /50x.html;
        location = /50x.html {
        }
    }
```

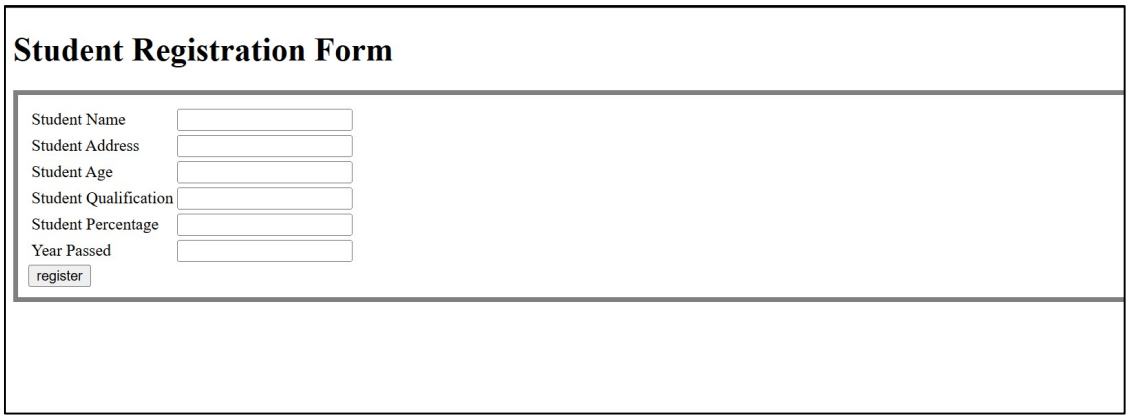
In the location, we need to give the private IP Address of the Application server with the folder name to identify the folder and website.

Step 20: After this, restart the nginx.

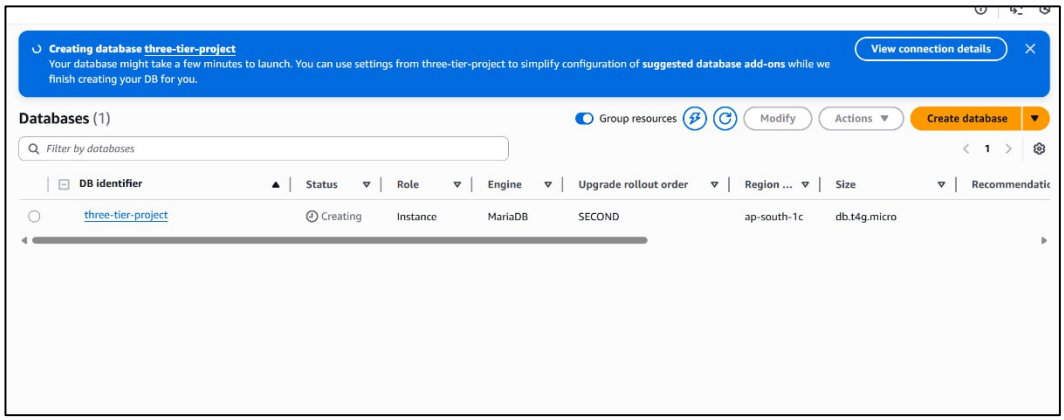
```
[root@ip-172-65-13-201 nginx]# vim nginx.conf
[root@ip-172-65-13-201 nginx]# systemctl restart nginx.service
[root@ip-172-65-13-201 nginx]#
```

Step 21: Then you can see the website after hitting the public IP address

Title: 3-Tier project (Tomcat & MySQL)

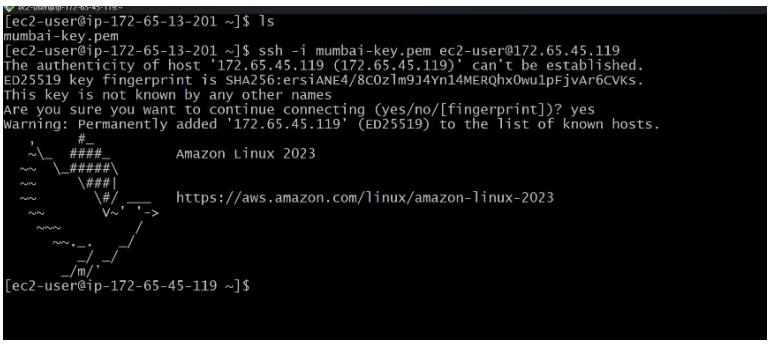


Step 22: So, to store the data, create an RDS Database

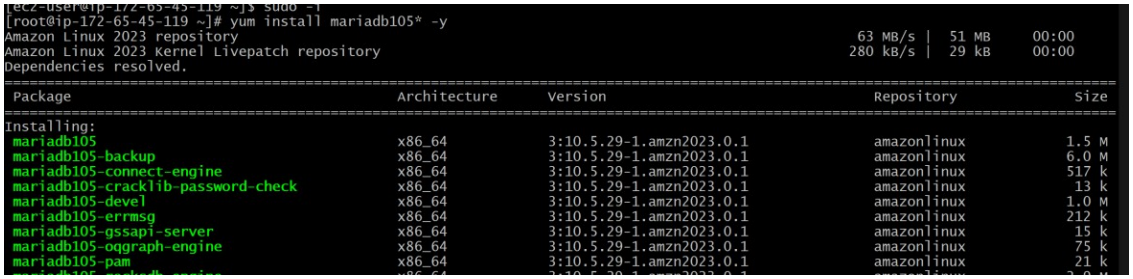


RDS creation select Mariadb – user name and manage password, and also select our VPC and security group. But to select the Availability zone, select the zone where you created db server and create an RDS DB instance there.

Step 23: Get access to the DB instance first with the key and db instance private IP address



Step 24: To access the RDS Database, we need to install mariadb105*.



Title: 3-Tier project (Tomcat & MySQL)

Now start it –

```
[root@ip-172-65-45-119 ~]# systemctl status mariadb.service
○ mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; preset: disabled)
     Active: inactive (dead)
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
[root@ip-172-65-45-119 ~]# systemctl start mariadb.service
[root@ip-172-65-45-119 ~]# systemctl enable mariadb.service
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[root@ip-172-65-45-119 ~]#
```

Step 24: After the creation of the RDS Database instance, you get an Endpoint with this, and you can access the RDS instance on the DB server. – mysql -h endpoint -u USERNAME -pPASSWORD

```
[root@ip-172-65-45-119 ~]# mysql -h three-tier-project.cfgsam2y0p3w.ap-south-1.rds.amazonaws.com -u arya -pArya12345$
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 72
Server version: 11.4.8-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create database studentapp;
Query OK, 1 row affected (0.004 sec)

MariaDB [(none)]> use studentapp;
Database changed
MariaDB [studentapp]> CREATE TABLE if not exists students(student_id INT NOT NULL AUTO_INCREMENT,
    -> student_name VARCHAR(100) NOT NULL,
    ->     student_addr VARCHAR(100) NOT NULL,
    -> student_age VARCHAR(3) NOT NULL,
    -> student_qual VARCHAR(20) NOT NULL,
    -> student_percent VARCHAR(10) NOT NULL,
    -> student_year_passed VARCHAR(10) NOT NULL,
    -> PRIMARY KEY (student_id)
    -> );
Query OK, 0 rows affected (0.018 sec)

MariaDB [studentapp]> select * from studentapp;
ERROR 1146 (42S02): Table 'studentapp.studentapp' doesn't exist
MariaDB [studentapp]> show students
    -> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the ri
ght syntax to use near 'students' at line 1
MariaDB [studentapp]> show tables;
+----------------------+
| Tables_in_studentapp |
+----------------------+
| students             |
+----------------------+
1 row in set (0.001 sec)
```

In this, also configure the Database and table for the website with commands.

Step 25: Now log out of the DB server and log in application server to connect application database.

After logging root of the app, go to - /opt/apache-tomcat-9.0.113/lib/

Here, we have to download **mysql connector** to connect to the Database.

```
[ec2-user@ip-172-65-23-9 apache-tomcat-9.0.113]$ sudo -i
[root@ip-172-65-23-9 ~]# cd /opt/apache-tomcat-9.0.113/
[root@ip-172-65-23-9 apache-tomcat-9.0.113]# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
[root@ip-172-65-23-9 apache-tomcat-9.0.113]# cd lib/
[root@ip-172-65-23-9 lib]# ls
annotations-api.jar       catalina.jar    jsp-api.jar            tomcat-i18n-cs.jar  tomcat-i18n-pt-BR.jar  tomcat-util.jar
catalina-ant.jar          ecj-4.20.jar    servlet-api.jar        tomcat-i18n-de.jar  tomcat-i18n-ru.jar     tomcat-websocket.jar
catalina-ha.jar           el-api.jar      tomcat-api.jar         tomcat-i18n-es.jar  tomcat-i18n-zh-CN.jar  websocket-api.jar
catalina-ssi.jar          jasper-el.jar   tomcat-coyote-ffm.jar  tomcat-i18n-fr.jar  tomcat-jdbc.jar
catalina-storeconfig.jar  jasper.jar      tomcat-coyote.jar      tomcat-i18n-ja.jar  tomcat-jni.jar
catalina-tribes.jar       jaspic-api.jar  tomcat-dbcp.jar        tomcat-i18n-ko.jar  tomcat-util-scan.jar
[root@ip-172-65-23-9 lib]# curl -O https://s3-us-west-2.amazonaws.com/studentapi-cit/mysql-connector.jar
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  983k  100  983k    0     0   460k      0  0:00:02  0:00:02 --:--:--  460k
[root@ip-172-65-23-9 lib]# |
```

Step 26: Now we have to add the location of db in the application server, so go to - /opt/apache-tomcat-9.0.113/conf/

Here, you see **content.xml,** so in this, we add the location and save it.

```
[root@ip-172-65-23-9 lib]# cd ..
[root@ip-172-65-23-9 apache-tomcat-9.0.113]# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
[root@ip-172-65-23-9 apache-tomcat-9.0.113]# cd conf
[root@ip-172-65-23-9 conf]# ls
Catalina          catalina.properties  jaspic-providers.xml  logging.properties  tomcat-users.xml  web.xml
catalina.policy   context.xml          jaspic-providers.xsd  server.xml          tomcat-users.xsd
[root@ip-172-65-23-9 conf]# vim context.xml |
```

Title: 3-Tier project (Tomcat & MySQL)

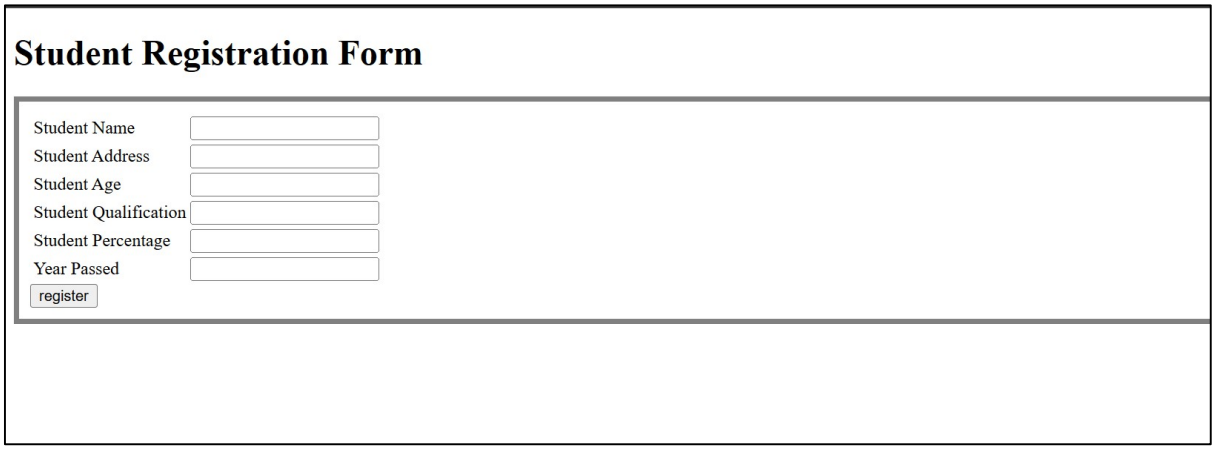
```
    <!-- Default set of monitored resources. If one of these changes, the    -->
    <!-- web application will be reloaded.                                   -->
    <WatchedResource>WEB-INF/web.xml</WatchedResource>
    <WatchedResource>WEB-INF/tomcat-web.xml</WatchedResource>
    <WatchedResource>${catalina.base}/conf/web.xml</WatchedResource>

    <!-- Uncomment this to disable session persistence across Tomcat restarts -->
    <!--
    <Manager pathname="" />
    -->
    <Resource name="jdbc/TestDB" auth="Container" type="javax.sql.DataSource"
    maxTotal="500" maxIdle="30" maxWaitMillis="1000"
    username="arya" password="Arya12345$" driverClassName="com.mysql.jdbc.Driver"
    url="jdbc:mysql://three-tier-project.cfgsam2yOp3w.ap-south-1.rds.amazonaws.com:3306/studentapp?useUnicode=yes&amp;characterEncoding=utf8"/>
</Context>
~
~
~
~
~
```

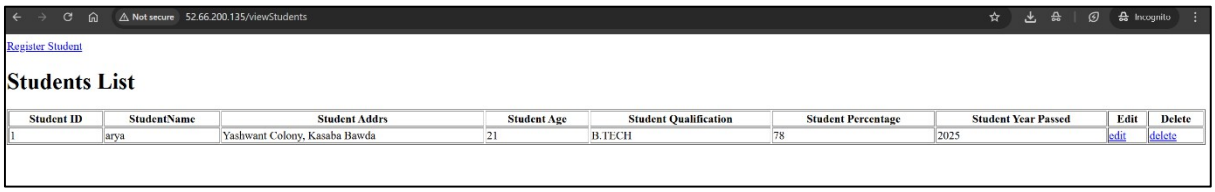
In this, we give the username and password of the database and also the endpoint of the RDS Instance.

Step 27: Then stop the Tomcat and start it again.

```
[root@ip-172-65-23-9 conf]# cd ..
[root@ip-172-65-23-9 apache-tomcat-9.0.113]# cd bin
[root@ip-172-65-23-9 bin]# ls
bootstrap.jar       ciphers.bat                    configtest.bat  digest.sh         setclasspath.sh  startup.sh            tool-wrapper.sh
catalina-tasks.xml  ciphers.sh                     configtest.sh   makebase.bat      shutdown.bat     tomcat-juli.jar       version.bat
catalina.bat        commons-daemon-native.tar.gz   daemon.sh       makebase.sh       shutdown.sh      tomcat-native.tar.gz  version.sh
catalina.sh         commons-daemon.jar             digest.bat      setclasspath.bat  startup.bat      tool-wrapper.bat
[root@ip-172-65-23-9 bin]# ./catalina.sh restart
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.113
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.113
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.113/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.113/bin/bootstrap.jar:/opt/apache-tomcat-9.0.113/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Usage: catalina.sh ( commands ... )
commands:
  debug             Start Catalina in a debugger
  debug -security   Debug Catalina with a security manager
  jpda start        Start Catalina under JPDA debugger
  run               Start Catalina in the current window
  run -security     Start in the current window with security manager
  start             Start Catalina in a separate window
  start -security   Start in a separate window with security manager
  stop              Stop Catalina, waiting up to 5 seconds for the process to end
  stop n            Stop Catalina, waiting up to n seconds for the process to end
  stop -force       Stop Catalina, wait up to 5 seconds and then use kill -KILL if still running
  stop n -force     Stop Catalina, wait up to n seconds and then use kill -KILL if still running
  configtest        Run a basic syntax check on server.xml - check exit code for result
  version           What version of tomcat are you running?
Note: Waiting for the process to end and use of the -force option require that $CATALINA_PID is defined
[root@ip-172-65-23-9 bin]# ./catalina.sh stop
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.113
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.113
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.113/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.113/bin/bootstrap.jar:/opt/apache-tomcat-9.0.113/bin/tomcat-juli.jar
Using CATALINA_OPTS:
NOTE: Picked up JDK_JAVA_OPTIONS:  --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.lang.invoke=ALL-UNNAMED --add-opens=java.b
ase/java.lang.reflect=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=java.base/java.u
til.concurrent=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED
[root@ip-172-65-23-9 bin]# ./catalina.sh start
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.113
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.113
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.113/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.113/bin/bootstrap.jar:/opt/apache-tomcat-9.0.113/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
```

After all these steps, we create a 3-Tier project.





Here you can see where all the data will be stored.